IN PURSUIT OF ETHICAL AI

Always walking a fine line

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Eighty years ago in 1942, acclaimed science fiction writer, Isaac Asimov had the foresight to anticipate the potential risks of autonomous, near-sentient robots or technology systems of the future to propose his Three Laws of Robotics in order to limit their danger to humanity. In Asimov's fictional ethical code, robots would never harm humans (1st law), robots would always obey humans (2nd law), and robots would always protect themselves (3rd law) as long as they didn't contradict the first two laws.

Asimov understood that technological advancement was inevitable, the problem he wanted to avoid was the unintended consequences of poorly designed technology. By proposing his Three Laws of Robotics, he was undoubtedly forcing machines to operate under a moral code of conduct. If you expand that logic to AI, then it's nothing but forcing the architects of game-changing AI applications to take greater moral responsibility for their actions – from the drawing board itself, not after releasing a poorly thought-out or potentially dangerous AI algorithm to cause irreparable damage to our way of life.

"AI Ethics can be defined as a socio-technical lens on the design and impact of AI solutions on our societies, involving the application of principles and techniques to ensure responsible development and use of AI technologies," according to Ria Cheruvu, Lead Architect - AI Ethics, Intel. AI Ethics can cover various domains, she says, including Transparency; Sustainability; Fairness and Bias; and Security, Safety, and Privacy.

NEED FOR ETHICAL AI

The importance of ethical AI is highlighted from not only a technical or organizational standpoint, it also takes into account individual and societal perspectives around some serious topics: Algorithmic bias causing harm and discrimination to certain populations. Transparency and trust concerns with AI systems, with the potential to severely impact consumer confidence and brand reputation. Technical failures and vulnerability to attacks, particularly critical to assess for high-risk scenarios and data types, such as vulnerable and sensitive medical records. Impact of AI systems on the climate, such as the carbon footprint or AI systems, and even copyright and license concerns for AI technologies in the future, among other things.

RIA CHERUVU
Lead Architect - AI Ethics, Intel

Mariagrazia Squicciarini, Chief of Executive Office and Director AI, UNESCO, explains this further. "AI has an advantage that is much more pervasive – so it's not only one sector, it can help all sectors. And what you ultimately need is skilled human resources, good computational ability and access to data," she says.

Ethical AI means that it needs to be better framed for people to be able to not give their data by the time they don't want to, or know what they're used for and consent to it. Ethical AI can level the playing field, allow developing countries to leapfrog ahead of advanced economies to benefit from technological progress, according to Squicciarini.

"At the levels of organizations and businesses, Ethical AI can be approached from defining principles and guidelines to ensure responsible development practices are followed within an organization. These processes and objectives are known as ethical governance," says Intel's Ria Cheruvu.

"At the technical level, toolkits, frameworks, and methodologies can play a key role in accelerating identifying issues with AI systems early-on (e.g., transparency mechanisms)